K J SOMAIYA INSTITUTE OF TECHNOLOGY
Department of Information Technology

Academic Year: 2026-27
Class: S.Y. B.Tech.
Semester: III

Subject: DSA LAB
Course Code: ITL302

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EXP NO:3 -INFIX TO POSTFIX

PROGRAM:

```
#include<stdio.h>
#include<conio.h>
#include<ctype.h>
#include<string.h>
#define MAX 100
char stk[MAX];
int top=-1;
void push(char ch)
{
 if(top==MAX-1)
 printf("overflow\n");
 else
 st[top++]=ch;
}
char pop()
{
if(top==-1)
return -1;
else
return stk[top--];
}
```

```
int priority(char ch)
{
 if(ch=='^')
 return 3;
 else if (ch=='*'|| ch=='/')
 return 2;
 else if (ch=='+'|| ch=='-')
 return 1;
 else
 return 0;
}
void infixtopostfix(char infix[])
{
char postfix[MAX];
int i,j=0;
char ch;

for(i=0;infix[i]!='\0';i++)
{ch=infix[i];
if(isalnum(ch))
{
for(i=0;infix[i]!='\0';i++)
{ch=infix[i];
if(isalnum(ch))
{
postfix[j++]=ch;
}
else if(ch=='(')
{
 push(ch);
}
```

```
else if (ch == ')')
{
while (top != -1 && stack[top] != '(')
{
postfix[j++] = pop();
}
pop();
}
else
{
while (top != -1 &&
stack[top] != '(' &&
priority(stack[top]) >= priority(ch))
{
postfix[j++] = pop();
}
push(ch);
}
}
while (top != -1)
{
```

```
}
while (top != -1)
{
postfix[j++] = pop();
}
postfix[j] = '\0';
printf("\nPostfix Expression = %s", postfix);
}

void main()
{
char infix[MAX];
clrscr();

printf("Enter Infix Expression: ");
scanf("%s", infix);

infixToPostfix(infix);

getch();
}
```

OUTPUT:

